Annex A: Task Profile Reference

Challenge 2: The Quantum Routing Brain

Schema documentation and usage guide for the 47-task CSV provided in the challenge repository (`task_profile.csv`).

PURPOSE

The task profile CSV is the knowledge base your routing engine consumes. Each row represents a canonical defense compute task that might appear in a tactical data stream. Your router reads the metadata columns to decide: run locally on classical hardware, offload to a remote classical accelerator, or send to a quantum backend.

The 47 tasks span seven categories covering the four mission threads in the challenge brief. You do not need to implement routing for all 47. Pick the tasks relevant to your chosen thread and use the rest as test cases to demonstrate your router handles diverse workloads.

SCHEMA REFERENCE

task_id — Unique identifier. Format: category prefix + sequential number (e.g., RF-07, SM-01, C2-04, CRYPTO-02, SAT-02).

task_name — Human-readable name of the compute task.

task_category — One of seven categories: RF Sensor Fusion (12 tasks), Spectrum Management (6), C-UAS Track Correlation (10), JADC2/C2 (12), Cryptography/Comms (4), Satellite Ops (3). Categories map to mission threads but are not exclusive to one thread.

computational_step — Brief description of what the task does within a mission workflow.

classical_algorithm — The standard classical approach. This is your baseline. Your router should default to this unless there is a defensible reason to send the task to a quantum backend.

typical_classical_runtime_ms — Approximate wall-clock time on representative tactical hardware (ARM Cortex-A78 class edge device or Xeon-class server). Order of magnitude, not a benchmark. Use for latency budget comparisons.

input_size_bytes — Typical input payload size. Relevant for bandwidth calculations when deciding whether to offload over a constrained tactical link.

output_size_bytes — Typical output payload size.

latency_budget_ms — Maximum allowable end-to-end latency for the task to be operationally useful. If the round trip to a remote QPU exceeds this budget, the router must keep the task local.

deadline_class — One of three categories: **hard-realtime** (miss the deadline and the result is useless or dangerous), **soft-realtime** (late results degrade but do not break the mission), **batch** (minutes to hours acceptable). Hard-realtime tasks should almost never be offloaded to a remote QPU.

classification_level — One of: UNCLASS, CUI, SECRET, TS-SCI. Commercial QPUs operate at most at IL5 (CUI). Any task classified SECRET or above cannot be offloaded to a commercial quantum backend. This is a hard constraint your router must enforce.

priority — 1 (highest) to 3 (lowest). Priority 1 tasks get first claim on local compute resources and should not be delayed by quantum offload queue times.

network_required — Where the task can execute: **local** (must run on the edge device), **edge-LAN** (can run on a nearby server within the tactical enclave), **SATCOM-OK** (can tolerate the latency of a satellite backhaul to rear-area compute), **DDIL-tolerant** (can tolerate intermittent connectivity and delayed results).

quantum_candidate — Whether the task is a plausible candidate for quantum acceleration: **Y** (published research demonstrates quantum formulation), **maybe** (theoretical quantum formulation exists but no competitive results), **N** (no known quantum advantage, classical is definitively better). Only tasks marked Y or maybe should be considered for quantum routing.

quantum_algorithm — The specific quantum approach, if applicable: QAOA, QUBO (via quantum annealing), QSVM/quantum kernel, HHL (linear systems), VQE, QRNG, or none.

expected_quantum_speedup — Current state of quantum advantage for this task: **proven** (operationally validated, only QRNG today), **unproven** (research demonstrations exist but no wall-clock advantage over classical at operational scale), **unproven (dequantized)** (quantum approach has been matched or beaten by new classical methods), **unproven (FT-required)** (requires fault-tolerant quantum hardware not yet available), **none** (no quantum speedup expected).

error_tolerance — What quality of quantum result is acceptable: **NISQ-tolerant** (approximate/noisy results are useful), **FT-required** (exact results needed, requires error-corrected hardware), **hybrid-OK** (quantum result used as initialization for classical refinement), **n/a** (not a quantum candidate).

TASK CATEGORIES AT A GLANCE

Category	Tasks	Quantum	Key Insight
RF Sensor Fusion	12	4 maybe	Most RF signal processing is hard-realtime with sub-10ms budgets. Only higher-level classification and geolocation tasks have latency budgets and problem structures amenable to quantum.
Spectrum Management	6	3 Y/maybe	Frequency assignment (graph coloring) is the canonical QAOA demo problem. But classical ILP solvers dominate at operational scale.
C-UAS Track Correlation	10	3 maybe	Kalman filters and particle filters are definitively classical. The combinatorial assignment sub-problems in MHT, JPDA, and Hungarian are the quantum-eligible components.
JADC2 / C2	12	4 Y/maybe	VRP, weapon-target assignment, and mission scheduling are the strongest quantum research candidates. But most are SECRET-classified and batch-tolerant, meaning the quantum offload path must handle both constraints.
Cryptography / Comms	4	1 Y	QRNG is the only operationally deployed quantum technology in this entire CSV. AES, PQC, and hashing are definitively classical.
Satellite Ops	3	2 Y/maybe	Constellation scheduling is a natural QUBO. Collision avoidance is theoretically VQE-eligible but practically FT-required.

THE HEADLINE NUMBER

> Across all 47 tasks: ∼**10 are credible NISQ research candidates** (QUBO assignment family + small-data quantum kernel classifiers). **Exactly 1 is operationally deployable on quantum hardware today** (QRNG). Roughly **35 of 47 tasks should route classical with high confidence**. Your routing engine's primary value is enforcing that default while leaving an instrumented, auditable path for the research candidates.

HOW TO USE THIS CSV

For Your Router

Load the CSV as your routing engine's decision table. At minimum, your router should check three columns before offloading any task to a quantum backend:

1. `classification_level`: Is the task UNCLASS or CUI? If SECRET or TS-SCI, reject the offload immediately.
2. `latency_budget_ms` vs. estimated round-trip time: Can the task tolerate the network latency to the QPU plus queue time plus execution time? If not, run classical locally.
3. `quantum_candidate`: Is the value Y or maybe? If N, do not waste QPU cycles.

After these three gates, secondary factors apply: `deadline_class` (hard-realtime tasks should almost never offload), `priority` (priority 1 stays local), `network_required` (if local-only, offload is architecturally impossible), and `expected_quantum_speedup` (if dequantized or FT-required, classical is the correct choice in 2026).

For Your Demo

Pick 5–10 tasks relevant to your chosen mission thread. Show them flowing through the router in real time. The compelling demo moment is not the quantum execution itself. It is the router correctly *refusing* to offload most tasks (because they are hard-realtime, or SECRET, or not quantum-eligible) and selectively sending 1–2 tasks to the quantum backend with full justification logged.

For Your Venture Pitch

The ratio matters. If your router sends everything to quantum, you have not understood the problem. If your router sends nothing, you have not demonstrated value. The right answer in 2026 is roughly 2–5 of 47 tasks are quantum-routable under favorable conditions, and even those should run alongside a classical baseline for validation. That is the honest pitch: the orchestration layer's value is managing this transition intelligently, not pretending quantum is ready for everything.

KEY REFERENCES FOR TASK DEFINITIONS

Source	Covers
JDL/Steinberg-Bowman Data Fusion Process Model	Levels 0–5 of data fusion used to categorize task complexity. Level 0 (signal processing) through Level 5 (user refinement).
Hall & Llinas, *Handbook of Multisensor Data Fusion*	Canonical reference for sensor fusion task definitions, algorithms, and architectures.
RAND RR-4408z1 (Lingel et al., 2020)	JADC2 AI/ML framework. Defines compute task categories and latency classes for joint command and control.
RAND RR-A263-1 (Walsh et al., 2021)	ML-assisted C2 decision support. Task decomposition for AI-augmented command workflows.
Kessel Run product lines (public)	Real-world Air Force software factory task categories. `https://kesselrun.af.mil/product-lines/`
Stoöß et al., IEEE Fusion 2021	Gate-based QAOA for multi-target tracking assignment. arXiv:2105.02011.
Choi & Kim, Eng. Appl. AI 2023	QAOA weapon-target assignment on IBM Lima hardware.
Wu et al., arXiv:2312.07821 (CSIRO Data61)	Quantum variational classifier for RF modulation classification.
Park et al., Springer 2024	QSVM for modulation classification on IonQ Harmony.
Bowles, Ahmed, Schuld (Xanadu, 2024)	Benchmarked 12 QML models on 160 datasets; classical baselines won nearly every case. arXiv:2403.07059.